Shubham Kumar

shubhamkumar.ethz@gmail.com
shubhamkriitr shubhamkumar101
Zürich, 8045, Switzerland *D.O.B.* 04-Nov-1996

Summary

Innovative and results-driven Software and Machine Learning Engineer with an MSc in Computer Science from ETH Zürich and over 4 years of industry experience, building and deploying scalable software and intelligent systems at Ucentrics AG, Disney Research, and Oracle. A versatile expert who combines strong full-stack software development skills with applied AI to create robust, production-ready applications.

Education

Master of Science, Computer Science — **ETH Zürich**
CGPA: 5.5/6 | *Major: Machine Learning, Minor: Data Management* — *September 2021 - June 2024*
Selected Coursework: Big Data, Information Security, Natural Language Processing for Law and Social Science, Advanced Machine Learning, Computational Intelligence

Bachelor of Technology, Electrical Engineering — **IIT Roorkee**
CGPA: 9.33/10 — *July 2014 - May 2018*
Selected Coursework: Design and Analysis of Algorithms, Programming in C++, Operating Systems

Work Experience

Ucentrics AG — **Zürich, Switzerland**
Machine Learning Engineer — *September 2024 - December 2025*
- Created an **automated ML training pipeline** and an inference pipeline optimized for low-compute devices from scratch, enabling end-to-end model development and deployment for 3D overlay. (Docker / Python / Google Cloud / ONNX)
- Created a Document Layout Analysis and **LLM** based tool for converting technical manuals into **digital workflows**, capable of digitizing 50-page documents within 3 minutes. (Python / Azure / LLMs / Prompt Engineering)
- Enhanced Ucentrics' core software with a **modular integration layer**, allowing seamless addition, replacement, debugging and configuration of AI models to power real-time scene understanding and visual inspection (C# / Unity / C++).
- Built a **synthetic data generation system** that automated the creation of 5 data modalities (RGB, depth, segmentation, landmarks, 6-DoF pose) for CAD models, cutting labeling effort by 80%. (Blender / Python)

ETH Zürich, Product Development Group — **Zürich, Switzerland**
Research Assistant — *December 2023 - May 2024*
- Engineered and deployed highly efficient image classification models, optimizing them for real-time mobile inference, improving processing speed by 35%. (C# / Python / C++)
- Developed Unity/Android **plugins**, enabling seamless voice-driven AR interactions and improving overall user engagement. (C# / Java / C++)

Disney Research | Studios — **Zürich, Switzerland**
Master's Thesis Student (Design and Training of Efficient Transformer Networks) — *March 2023 - September 2023*
- Created a tool for large-scale architecture search, experiment tracking, and reporting (**500+ runs**). (Python / FastAPI)
- Developed a network with **linear** computational & memory complexity in the number of tokens (points), reducing the mean reconstruction error of 3D-hand shapes by 10.3% while being **31.5% more computationally efficient.** (PyTorch)

Oracle Cloud AI Services — **Bengaluru, India**
Member of Technical Staff — *October 2020 - September 2021*
- Developed, trained, and productionized **Deep Learning** models for aspect-based sentiment analysis (ABSA) service, ensuring efficient implementation and high prediction quality (F1 **86.8%**). (PyTorch / ONNX / C++)
- Created **validation and benchmarking** agent, allowing easy registration of datasets, tasks, and metrics to generate customized and detailed performance summaries of the models for model selection. (Flask / Python)
- Created template processor for **synthetic text data** generation and pipeline for supporting **data annotation** and conflict resolution, which enabled a **4.5** times increase in high-quality in-house annotated data volume. (Python / Doccano)

Oracle Applications Lab **Bengaluru, India**
Software Engineer *July 2018 - September 2020*

- Designed and implemented **data aggregation API** for the supply chain dashboard, enabling the support team to quickly identify and resolve customer issues, improving turnaround time by **30%**. (Java / Spring Boot / JDBC)
- Implemented **backend and intent resolution flow** for a chatbot application which enabled users to **quickly query** shipping details via messengers like **Slack & Facebook**. (Node.js)
- Created a **web service** for dynamic pricing of pre-install software based on configurable rules. (Java / Spring Boot)
- Developed and maintained **Jenkins** pipelines to automate CI/CD workflows, improving reliability and deployment efficiency for Pre-install and OneView application components.
- Actively contributed to the requirement-gathering process and formulating **development plans** in collaboration with other teams (Costing, Inventory, and B2B).

Research Publications

***InfiNet: Fully Convolutional Networks for Infant Brain MRI Segmentation**, 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*, Washington, DC, 2018, pp. 145-148.

***Learning Optimal Deep Projection of ^{18}F-FDG PET Imaging for Early Differential Diagnosis of Parkinsonian Syndromes**, Deep Learning in Medical Image Analysis 2018, ML-CDS 2018, Granada, Spain.*

Technical Skills

Programming Languages:	Python, C#, C++, Java, JavaScript
Libraries/Frameworks:	PyTorch, Numpy, scikit-learn, PyBind, pytest, PySpark | FastAPI, Flask, Gunicorn, Spring Boot | NLTK, spaCy, Hugging Face, LangChain
Tools & Technologies:	Docker, Git, ONNX, Slurm, Jenkins, Grafana, PostgreSQL, Oracle
Software Development:	Design Patterns, Unit Testing, CI/CD
AI:	LLM, MCP, Transformers, CNN, VAE, GAN, Normalizing Flows, Linear Algebra
Certifications :	Scaled Agile Framework (SAFe 4) Practitioner

Selected Projects

Extracting Respiratory Rate with Uncertainty Estimates from PPG Using Probabilistic Deep Neural Networks
SIPLAB, ETH Zürich *October 2022 - December 2022*
Utilized **contrastive learning** to leverage unannotated signals, and developed a Deep Learning model in 'Mean Variance Estimation' setting, improving the MAE on the BIDMC respiratory dataset by **21%** (0.74 *breaths/min*) compared to the previous method, and also providing a **reliability estimate**.

PubMed Document Analysis
Machine Learning for Healthcare, ETH Zürich *April 2022*
Processed PubMed 200k RCT corpus, trained a Word2Vec embedding model, and used this along with a fully-connected network for classifying medical texts, achieving an F1 score of **84.29%**.

Road Segmentation
Computational Intelligence Lab, ETH Zürich *March 2022 - April 2022*
Developed a ResNet backbone-based aerial image segmentation pipeline for delineating road and background pixels, achieving an F1 of **89.9%**.

References

- Dr. Julian Wolf, CTO, Ucentrics AG, Zürich : julian.wolf@ucentrics.com
- Dr. Tunç O. Aydin, Research Scientist at Disney Research, Zürich : tunc@disneyresearch.com
- Mr. Nagaraj Bhat, Staff Machine Learning Engineer, ServiceNow, Bengaluru: nagaraj.bhat@servicenow.com
- Dr. Sailesh Conjeti, Global Product Lead - AI at Olympus Europa SE, Hamburg : sailesh.conjeti@olympus.com